

AI Engineer for Data Scientists Associate Certification Study Guide

Please use this study guide to create your certification self-study plan. We've included the objectives you should meet for each assessed competency, with links to relevant practice assessments.

Exam AIEDSA101: Data Management, Model development, Exploratory analysis (Python)

1.1 Perform standard data import, joining and aggregation tasks using Python

- Aggregate numeric, categorical variables and dates by groups using Python.
- Filter data based on different criteria using Python.

1.2 Assess data quality and perform validation tasks

- Identify and replace missing values using Python.
- Perform different types of data validation tasks (e.g. consistency, constraints, range validation, uniqueness) using Python.
- Identify and validate data types in a data set using Python.

Related Assessments

[Data Manipulation with Python](#)

2.1 Prepare data for modeling by implementing relevant transformations.

- Create new features from existing data (e.g. categories from continuous data, combining variables with external data) using Python.
- Explain the importance of splitting data and split data for training, testing, and validation using Python.
- Explain the importance of scaling data and implement scaling methods using Python.
- Transform data for modeling using Python.

2.2 Assess and define modeling approaches for deep learning and other machine learning problems

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- Describe fundamental deep learning concepts and architectures
- Recall the fundamental concepts of machine learning (ML) including supervised and unsupervised learning, and common algorithms
- Describe fundamental reinforcement learning concepts

2.3 Implement standard modeling approaches for deep learning and other machine learning problems.

- Identify appropriate data partitioning in open-source datasets to train ML and DL models
- Understand how to use libraries and models to search for optimal models and parameters

2.4 Use suitable methods to assess the performance of a model

- Analyze the performance of different ML and DL models using technical metrics

Related Assessments

Assessment to be released shortly

3.1 Calculate metrics to effectively report characteristics of data and relationships between features using Python (DSA)

- Calculate measures of center (e.g. mean, median, mode) for variables using Python.
- Calculate measures of spread (e.g. range, standard deviation, variance) for variables using Python.
- Calculate skewness for variables using Python.
- Calculate missingness for variables and explain its influence on reporting characteristics of data and relationships in Python.

3.2 Use data visualization tools to demonstrate characteristics of data

- Distinguish between different types of data visualizations (bar chart, box plot, line graph, and histogram) in demonstrating the characteristics of data.

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- Interpret data visualizations (bar chart, box plot, line graph, and histogram) and summarize the characteristics of the data.

3.3 Read and analyze data visualizations to represent the relationships between features

- Distinguish between different types of data visualizations (scatterplot, heatmap, and pivot table) in representing the relationships between features.
- Interpret the data visualizations (scatterplot, heatmap, and pivot table) and summarize the relationship between features.

Related Assessments

[Exploratory Analysis Theory](#)

Exam AIEDSA102: Data Management, Model development, Exploratory analysis (Python)

1.1 Explain ethics and bias concerns in relation to generative AI

1.2 Identify key privacy laws and themes surrounding data use, such as GDPR, CCPA and PIPL, and the themes of data sovereignty, data breach notifications, user data requests, increased transparency, accountability, and penalties for violating privacy law

Related Assessments

Assessment to be released shortly

2.1 Use common programming constructs to write repeatable production quality code for analysis in Python. (DSA)

- Define, write and execute functions
- Use and write the control flow statements
- Use and write loops and iterations

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2.2 Describe software development and deployment best practices and workflows and explain their benefits for AI systems.

- Explain how traditional programming best practices can be adapted to work with AI systems

Related Assessments

Assessment to be released shortly

3.1 Demonstrate awareness of basic concepts in MLOps and LLMOps

- Define the features that require tracking during the MLOps and LLMOps lifecycles (hyperparameters, model versions, ...)
- Define the metrics calculated to monitor performance in production

3.2 Prepare data for use in ETL and ELT pipelines

- Write functions to convert data to formats ready for ingestion by AI models

3.3 Create prototype systems and applications that utilize generative AI.

- Adapt provided code to import data from an API
- Choose the appropriate components and adapt them for the AI system
- Use prompt engineering techniques to customize generative AI models for different use cases
- Use open-source libraries and tools to plug into existing APIs for user-facing generative AI applications

Related Assessments

Assessment to be released shortly